

JOHNSTOWN-CAMBRIA COUNTY AP, PA, USA

WMO: 725127

Lat: 40.316N

Lon: 78.834W

Elev: 696

StdP: 93.24

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 04726

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.9	-14.4	-21.5	0.6	-15.2	-19.0	0.8	-13.0	12.6	-4.8	11.8	-5.1	5.9	280	0.539

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.7	28.6	20.7	27.4	20.0	26.2	19.3	22.3	26.3	21.4	25.1	20.7	24.3	4.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.1	17.1	24.4	20.2	16.2	23.5	19.4	15.4	22.7	68.8	26.4	65.4	24.9	62.8	24.4	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.0	9.5	8.5	-19.6	30.9	3.6	1.9	-22.1	32.3	-24.2	33.4	-26.2	34.5	-28.8	35.8
			-20.0	23.7	3.3	1.2	-22.3	24.5	-24.3	25.2	-26.1	25.9	-28.5	26.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.2	-3.6	-2.3	2.2	9.0	14.3	18.7	20.7	20.0	16.5	10.5	4.4	-0.7
	DBStd	9.99	6.94	6.39	6.36	5.62	4.74	3.46	2.76	2.71	3.88	5.01	5.54	5.84
	HDD10.0	1697	424	346	255	83	14	0	0	0	2	56	182	334
	HDD18.3	3578	680	576	501	282	141	38	11	15	80	247	418	590
	CDD10.0	1408	3	2	13	55	148	260	332	309	197	71	14	3
	CDD18.3	248	0	0	1	3	17	48	84	67	25	4	0	0
	CDH23.3	1181	0	0	1	18	92	232	417	292	119	10	0	0
	CDH26.7	181	0	0	0	1	8	34	76	46	16	1	0	0
Wind	WSAvg	4.1	5.2	5.1	4.8	4.6	3.8	3.5	3.2	2.9	3.2	4.0	4.5	4.9
Precipitation	PrecAvg	1071	82	64	95	96	102	101	102	95	93	76	84	83
	PrecMax	1441	142	99	157	199	160	154	179	217	235	146	229	169
	PrecMin	802	41	32	34	33	38	52	50	39	25	23	11	35
	PrecStd	164	31	22	35	41	40	29	33	42	53	38	42	35
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.1	17.0	21.4	25.9	27.7	29.3	30.7	30.0	28.7	24.8	19.7	16.2
		MCWB	12.1	11.3	13.5	15.4	18.8	21.0	21.8	21.2	19.6	18.1	13.1	12.1
	2%	DB	12.5	12.8	17.8	22.7	25.9	27.5	28.7	27.9	26.5	22.2	17.0	13.0
		MCWB	10.5	9.4	11.2	13.6	17.7	20.1	21.3	20.6	19.0	16.4	11.8	10.5
	5%	DB	10.0	9.7	14.9	20.4	23.9	26.2	27.3	26.4	24.5	20.1	14.6	10.7
		MCWB	8.0	6.5	9.9	12.6	16.8	19.6	20.3	19.9	18.2	15.5	10.6	8.3
	10%	DB	6.4	7.0	12.2	17.8	22.0	24.2	25.9	24.9	22.6	17.9	12.7	7.7
		MCWB	4.3	4.0	8.4	11.4	16.2	18.4	19.6	19.2	17.6	14.1	9.5	5.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.9	12.6	14.5	17.6	20.5	22.6	23.7	23.2	21.9	19.9	15.2	13.2
		MCDB	14.3	15.4	19.8	22.7	25.5	27.2	28.6	27.2	25.5	22.5	18.7	14.9
	2%	WB	11.0	10.0	12.6	15.4	19.2	21.2	22.4	22.0	20.6	18.1	13.1	11.0
		MCDB	12.3	12.3	15.8	19.7	23.5	25.7	26.7	25.5	24.0	20.9	15.5	12.7
	5%	WB	8.1	6.7	10.9	13.9	18.0	20.4	21.5	21.1	19.6	16.5	11.6	8.7
		MCDB	9.5	9.3	14.7	18.4	22.2	24.6	25.3	24.5	22.8	18.9	13.9	10.5
	10%	WB	4.5	4.1	8.3	12.5	16.9	19.7	20.7	20.2	18.6	14.7	9.7	5.4
		MCDB	6.3	6.6	12.0	16.8	20.7	23.4	24.2	23.5	21.2	17.6	12.3	7.4
Mean Daily Temperature Range	5% DB	MDBR	7.0	7.6	8.6	10.0	9.4	8.7	8.7	8.5	8.6	8.2	7.3	6.4
		MCDBR	9.0	11.1	11.6	12.2	11.2	10.2	10.0	9.8	10.5	10.2	9.8	9.7
		MCWBR	7.9	8.7	7.8	6.8	5.7	5.1	4.5	4.5	4.9	5.7	6.8	8.1
	5% WB	MCDBR	8.8	10.7	10.9	10.8	9.7	8.9	8.6	8.5	8.3	8.3	9.0	9.5
MCWBR		8.1	8.9	8.2	7.2	5.6	4.9	4.4	4.5	4.5	5.6	7.5	8.9	
Clear-Sky Solar Irradiance	taub	0.297	0.310	0.336	0.364	0.401	0.445	0.458	0.454	0.375	0.349	0.308	0.290	
	taud	2.444	2.380	2.397	2.355	2.257	2.159	2.138	2.133	2.342	2.421	2.519	2.502	
	Ebn at Noon	869	906	918	909	877	836	820	812	863	849	852	852	
	Edh at Noon	79	97	107	120	135	149	151	148	112	93	73	69	
All-Sky Solar Radiation	RadAvg	1.60	2.44	3.38	4.55	5.15	5.66	5.68	5.10	4.17	2.86	1.96	1.34	
	RadStd	0.14	0.32	0.30	0.42	0.49	0.47	0.36	0.30	0.49	0.31	0.23	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (38 neighbors)		+0.37	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+114	+53

Nomenclature: See separate page